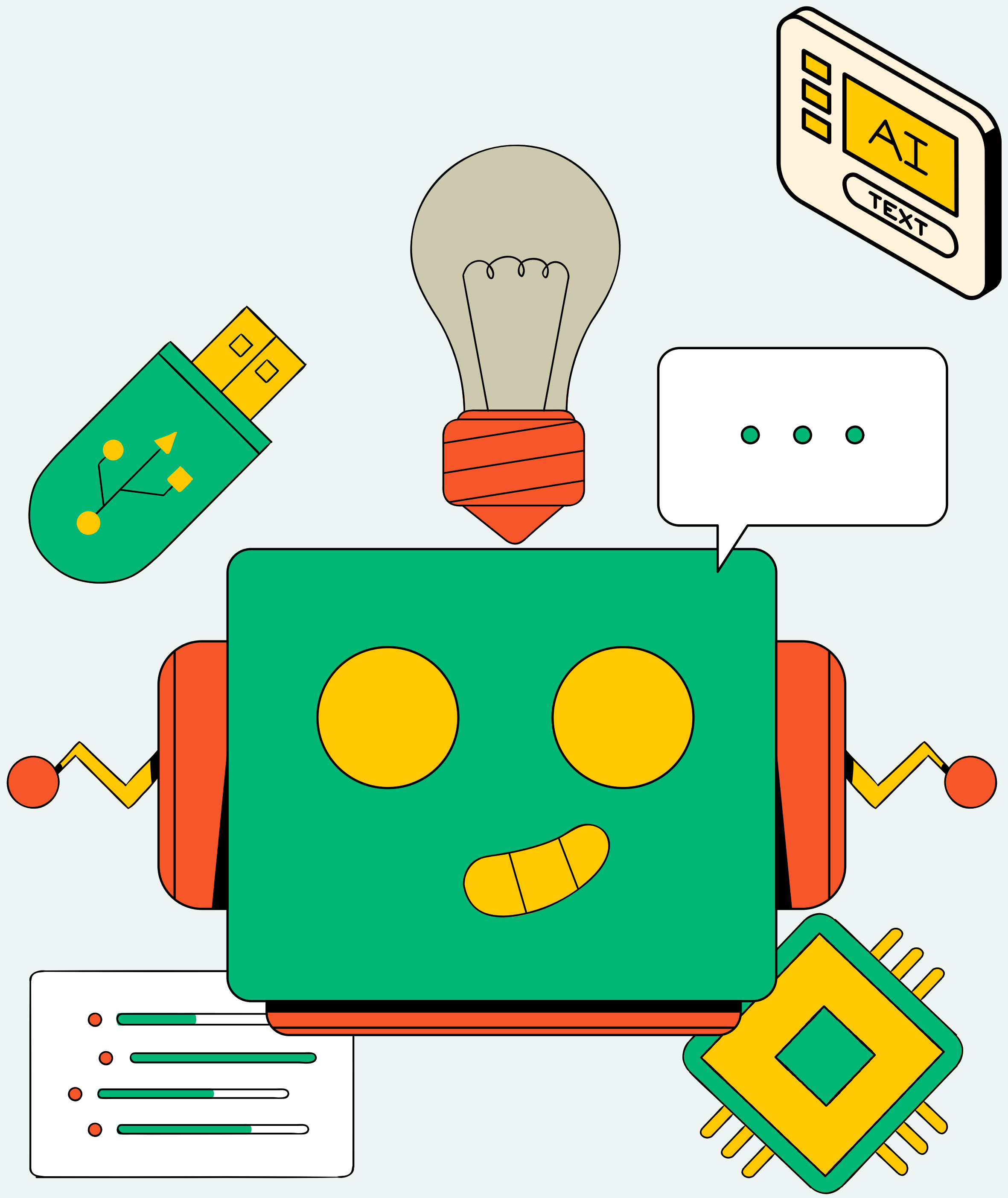
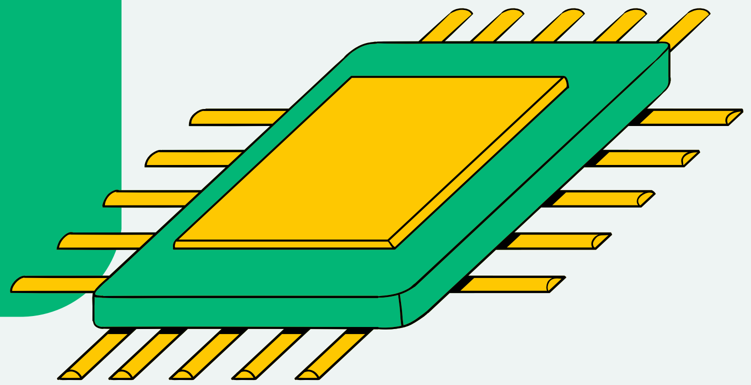
ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

PROPOSAL PRESENTATION



GROUP ID:

2026-Y2-S1-KU-01



TITLE

E-Commerce Customer Behavior Analysis and Segmentation

IT2011 - Artificial Intelligence and Machine Learning Group Proposal

Manathunge A.I IT23642096

Nayanajith K.A.T IT25100020

Ahamed M.N.A IT24104390

Herath H.M.A.R.B IT25100117

Sarma S S L IT25100002

Liyanage D.D.S IT25100119

ASSIGNED DATASET OVERVIEW

- **DATASET NAME:** ONLINE RETAIL II
- **SOURCE:** UCI MACHINE LEARNING REPOSITORY
- **DIRECTLINK:**
  https://archive.ics.uci.edu/dataset/502/online+retail+ii

- **DATASET CONTEXT:**
  - A REAL ONLINE RETAIL TRANSACTION DATASET.
  - REPRESENTS A UK-BASED, NON-STORE ONLINE RETAILER.
  - THE COMPANY PRIMARILY SELLS UNIQUE, ALL-OCCASION GIFT-WARE.
  - MANY CUSTOMERS ARE WHOLESALERS.

PROBLEM DOMAIN IDENTIFICATION

- **Domain:** Retail & E-Commerce Business Intelligence
- **The Real-World Problem:**
    - Online retailers struggle to identify their most valuable customers.
    - A "one-size-fits-all" marketing approach is costly and inefficient.

OUR PROPOSED AI SOLUTION

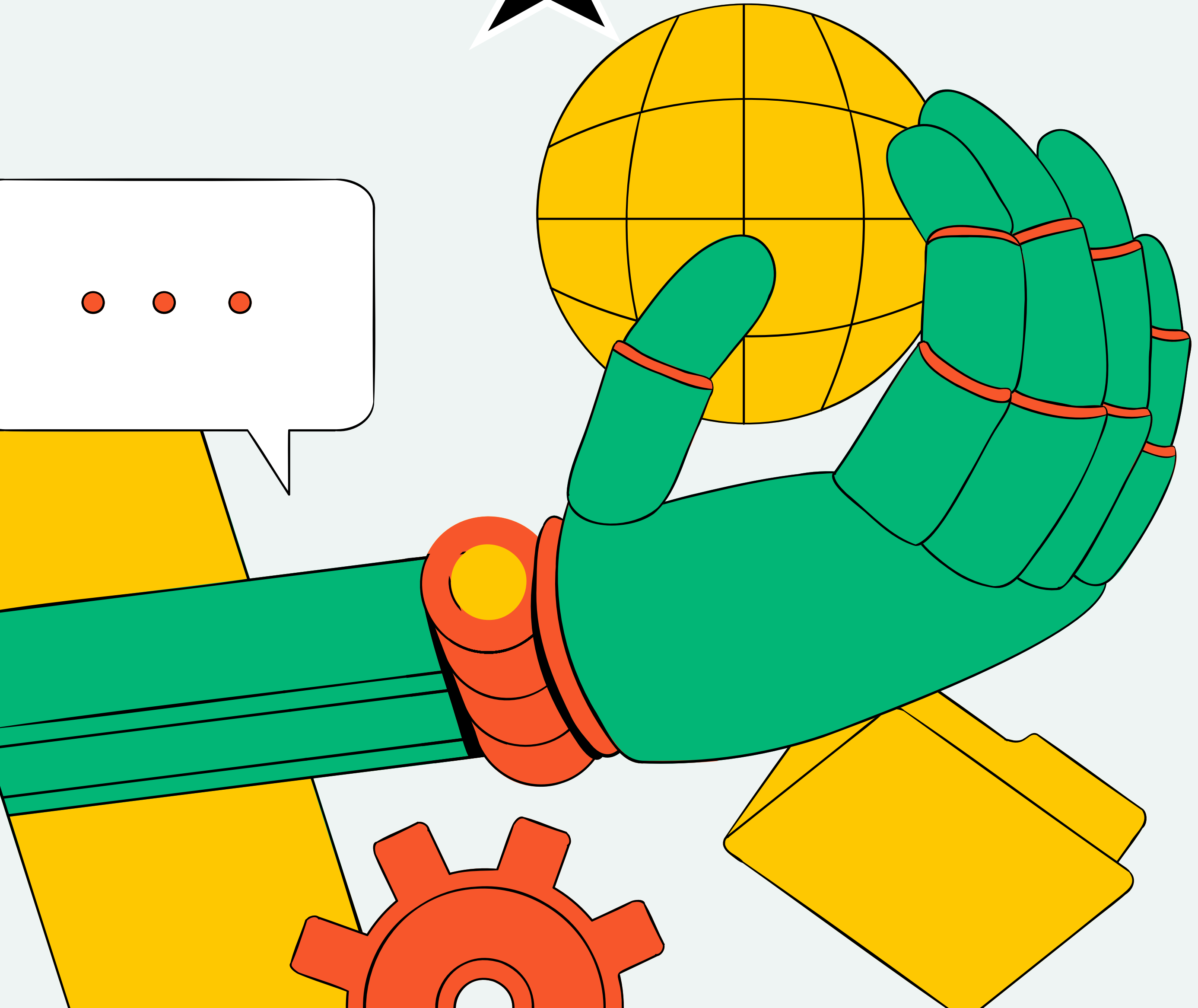
- Apply Unsupervised Machine Learning (Clustering).
- Group customers based on their buying behavior using RFM Analysis (Recency, Frequency, Monetary value).
- The generated RFM features are then analyzed using the K-Means Clustering algorithm.
- Enable the business to launch targeted, personalized marketing campaigns.

DATASET JUSTIFICATION - FEATURES & SUITABILITY

Instances
1067371

SIZE:

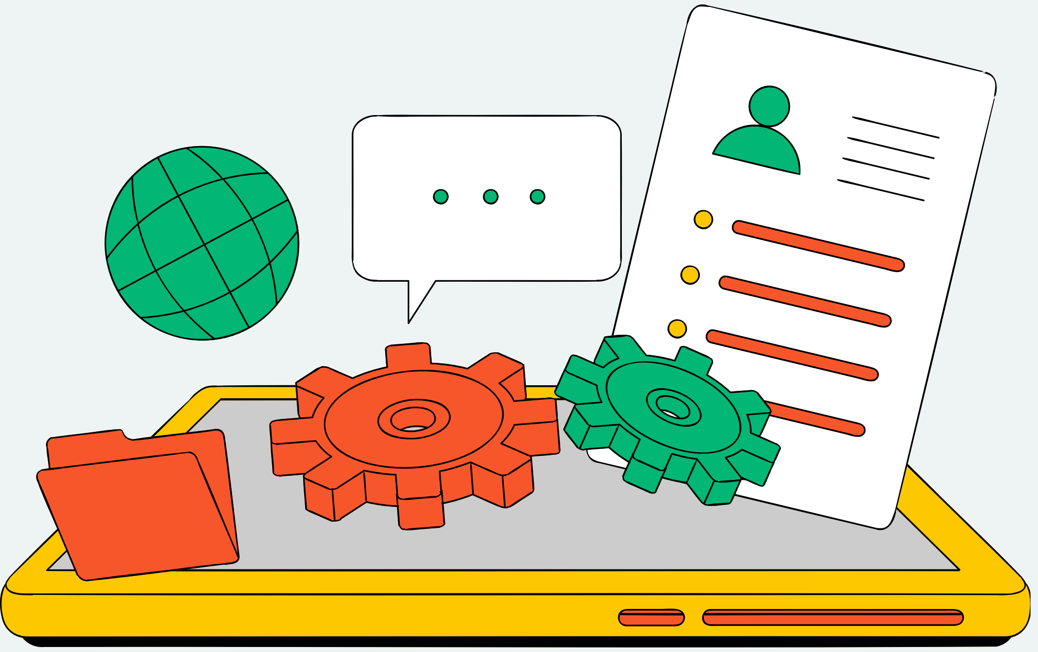
Over 1 million instances (transactions).



Key Features:

- **InvoiceNo & InvoiceDate:** Tracks when purchases happen (Recency & Frequency).
- **Quantity & UnitPrice:** Allows us to calculate total spend (Monetary Value).

- **CustomerID:** Essential for grouping transactions by individual buyers.
- **StockCode & Description:** Details about the items bought.
- **Country:** Geographic location of the customer.



- **Suitability:** Contains all necessary raw data to engineer the features required for our clustering models.

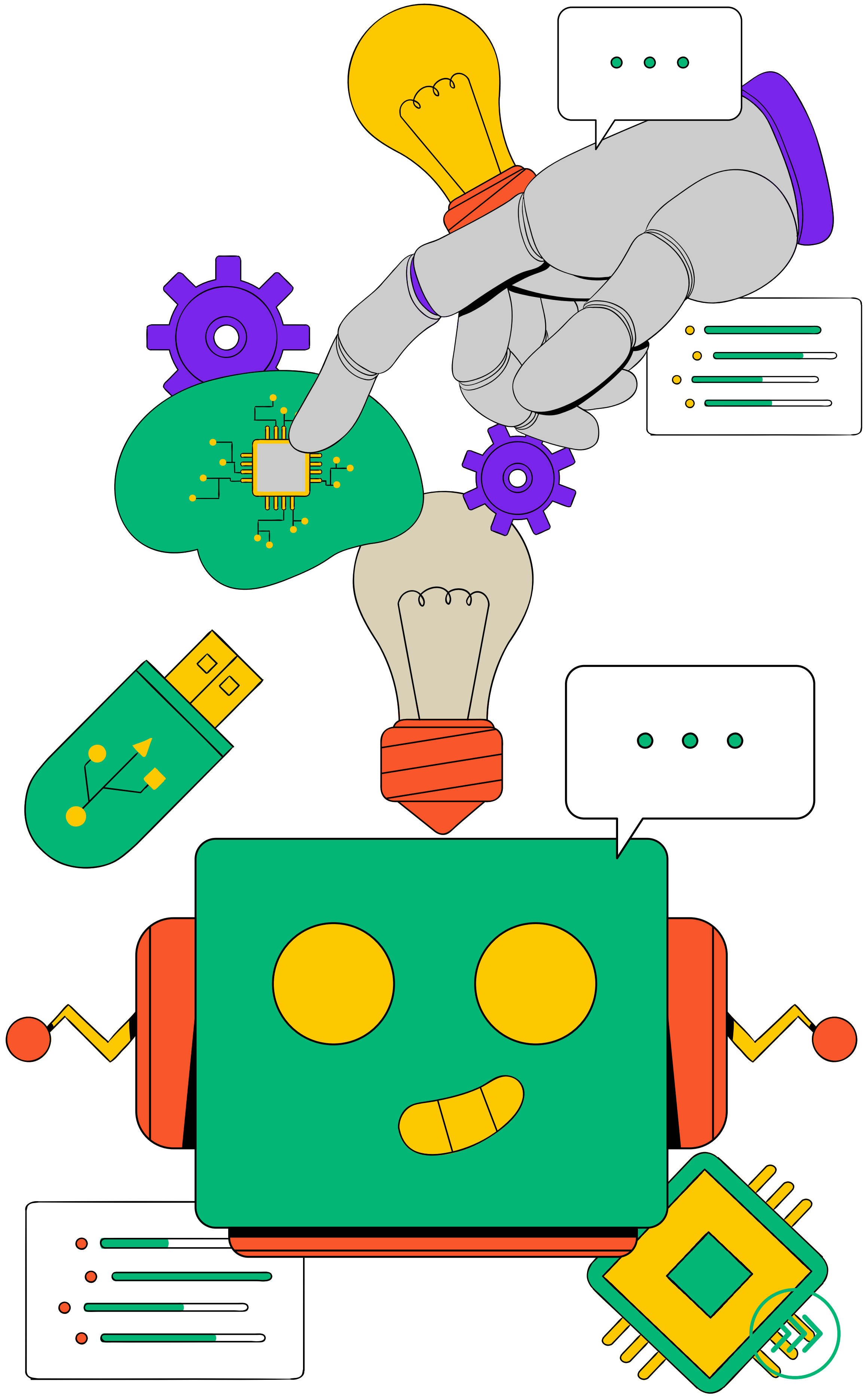
DATASET JUSTIFICATION - QUALITY & LIMITATIONS

Data Quality Challenges:

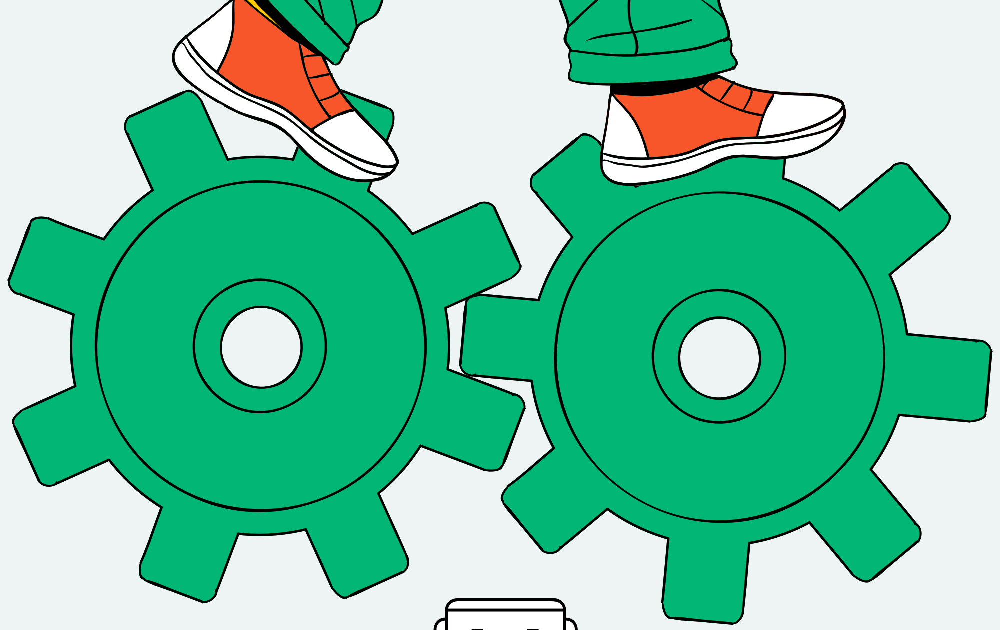
- **Missing Values:** Some transactions lack a CustomerID.
- **Anomalies / Outliers:** Cancellations are marked with a 'C' and often have negative quantities.

PLANNED MITIGATION

- We will perform extensive Data Preprocessing.
- Remove rows without CustomerIDs, as we cannot cluster unidentified users.
- Filter out cancelled transactions and handle negative anomalies.
- Normalize the data to ensure algorithms like K-Means perform optimally.



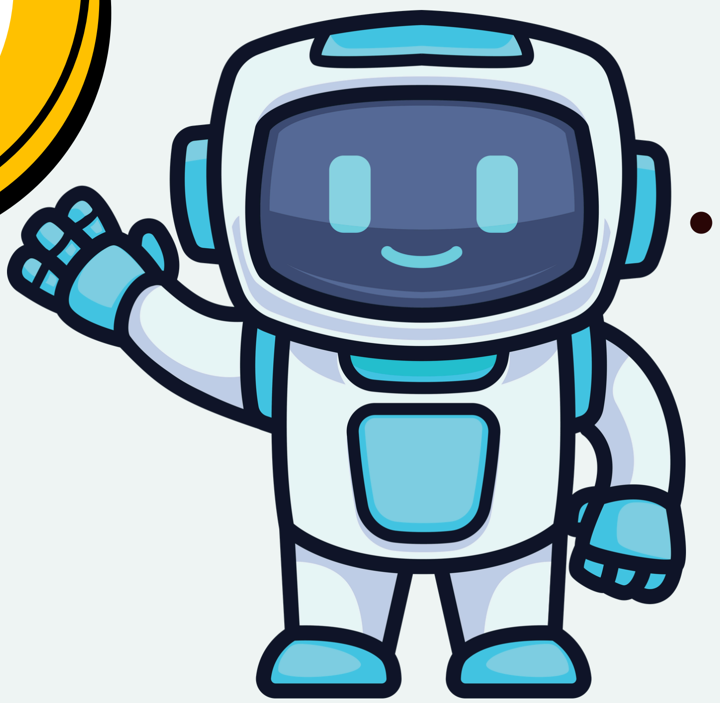
PLANNED AI TOOL USAGE & CONCLUSION



PRIMARY AI TOOL

Google Antigravity

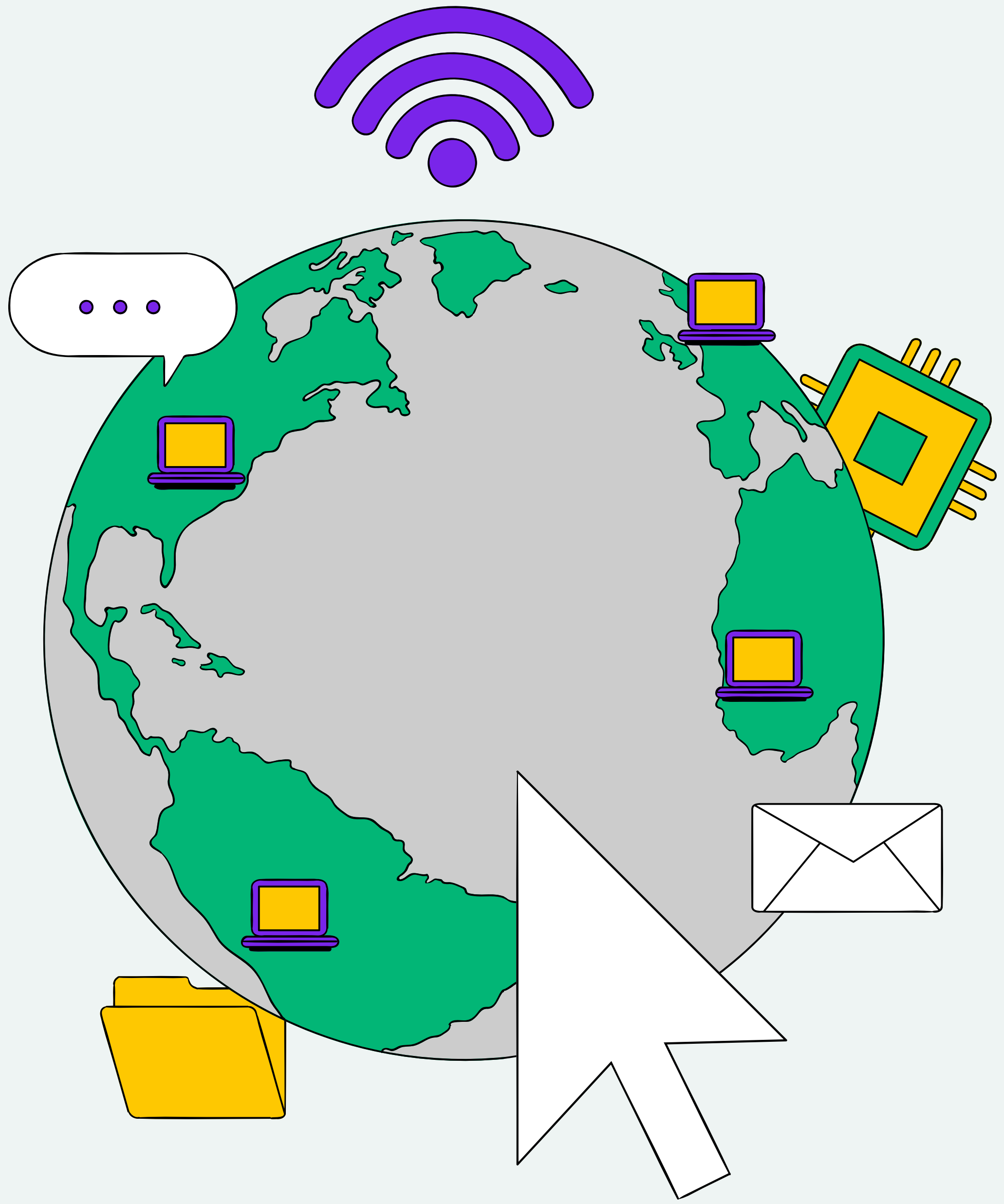
Advanced Agentic AI Coding Assistant



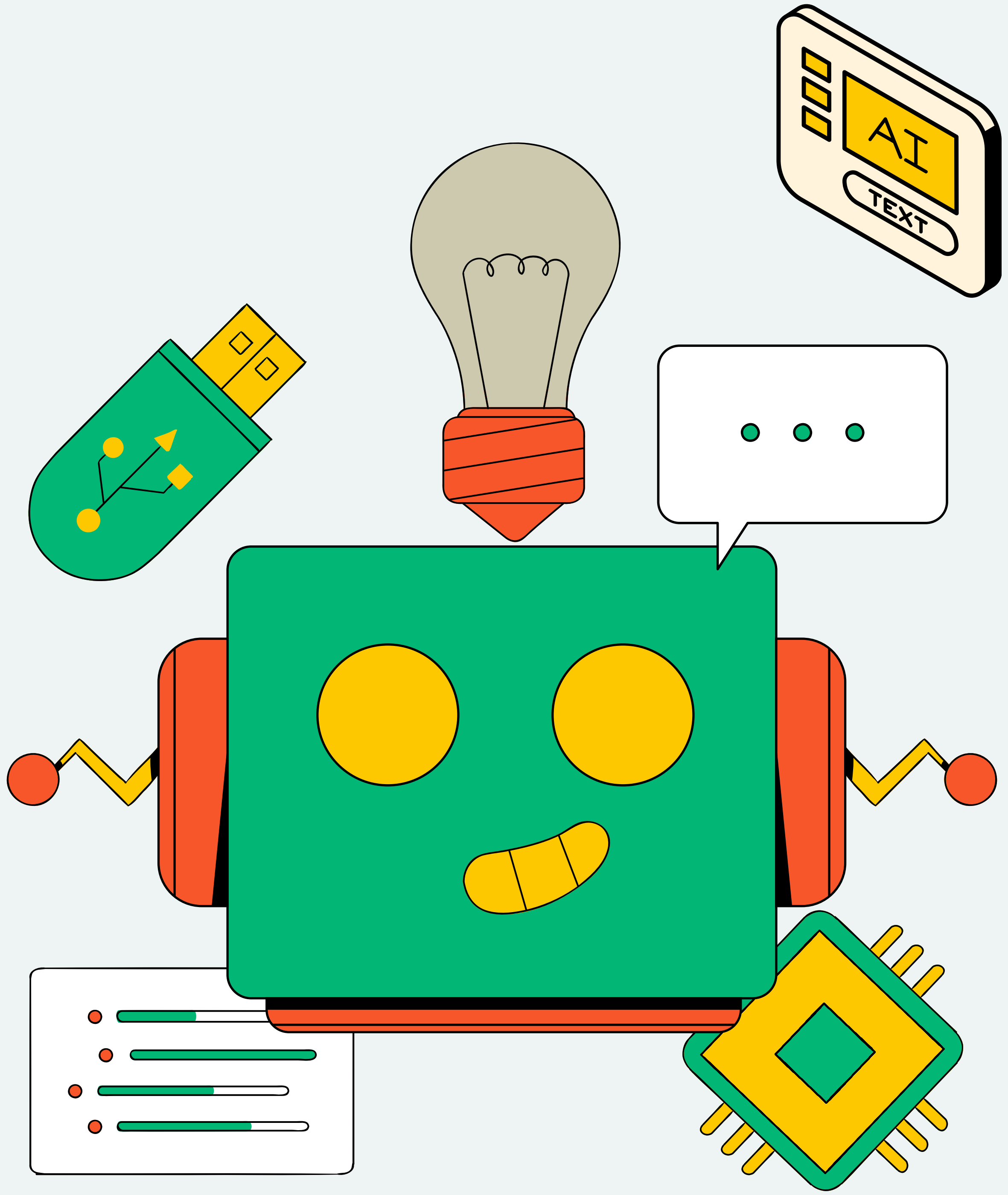
PLANNED TASKS FOR ANTIGRAVITY:

- **Code Generation & Debugging:** Writing Pandas and Scikit-learn boilerplate code.
- **EDA Ideation:** Brainstorming visualizations for our Exploratory Data Analysis.
- **Explanation & Brainstorming:** Helping us understand complex hyperparameter tuning concepts.

VERIFICATION PROCESS



- Our group will manually review, test, and take ownership of all AI-assisted code.
- Honest disclosure will be provided in the final AI Tool Usage Declaration section of our report.



QUESTIONS?

THANK YOU!